STL-50B2 Protocol Software Introduction

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Yes, the **STL-50B2** is a **DTOF LiDAR** from **LDROBOT**, with official protocol software and ROS driver packages available. Below is a ready-to-use solution:

1. PC Visualization Software (Windows)

Official Testing Tool: LdsPointCloudViewer (formerly ld_desktop)

This is LDROBOT's official desktop software for testing their STL-series LiDARs (including STL-50B2), supporting real-time point cloud visualization.

- Download: GitHub official release page
- Latest version: v2.3.13, compatible with Windows 10/11

Usage Steps:

1. Install the CP2102/CH340 serial driver (included with the adapter board).
2. Extract the software package and run `LdsPointCloudViewer.exe`.
3. Connect the LiDAR (proper wiring and power supply), select your COM port in the software, and set the baud rate to **115200**.
4. Select the LiDAR model (STL series), click **Connect**, and you will see 360° point cloud data.

Alternative General Tools

If the official software is unavailable, you can use compatible alternatives:

- YDLIDAR testing software: works with most serial 2D LiDARs (configure baud rate and frame format).
- Serial Assistant (SSCOM): verify LiDAR data output by checking continuous hexadecimal data streams.

2. ROS/ROS2 Driver Packages (for Raspberry Pi / Robots)

LDROBOT provides official ROS drivers for robot integration:

- ROS1 (Melodic/Noetic): `ldlidar_stl_ros`GitHub URL: Supports Ubuntu 18.04/20.04, compilable and runnable directly.
- ROS2 (Foxy/Humble): `ldlidar_stl_ros2`GitHub URL: Compatible with Raspberry Pi 4/5, usable after configuring serial port permissions.

3. Protocol & Data Format Description

The LiDAR uses LDROBOT's proprietary DTOF serial protocol. Core parameters:

- Communication: TTL UART (3.3V/5V compatible)
- Default baud rate: 115200 (8N1, no flow control)

- Data frame: starts with a fixed header, containing angle, distance, and signal strength data; each frame includes multiple ranging points.

The complete protocol parsing code is included in the driver package—**no need to write it yourself**.

4. Important Warnings

- Use **5V power supply only**; overvoltage will burn the mainboard.
- Raspberry Pi UART uses 3.3V logic level. A **level shifter** is required when connecting to the 5V LiDAR, otherwise the Raspberry Pi will be damaged.
- The baud rate in the software **must match** the LiDAR setting. Garbled data is usually caused by incorrect baud rate or wiring.